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Shubham Bhattacharya

AI Agent Backend Engineer — TypeScript, Node.js, Custom ReAct Loops

shubhambhattacharya107@gmail.com | +91-9155252394 | Vadodara, Gujarat, India

github.com/shubhambhattacharya-dev | linkedin.com/in/shubhambhattadev | shubhambhattacharya.dev

SUMMARY

AI Agent Backend Engineer specializing in production LLM systems, custom ReAct agent loops, and intelligent automation with TypeScript and Node.js. Built agentic backends using custom Zod-validated tool registries, multi-model fallback routing, and prompt injection protection — without LangChain abstractions. Strong focus on zero-trust AI architecture where every LLM output is validated before execution. Seeking AI Agent Backend Developer roles building production-grade agent systems.

TECHNICAL SKILLS

AI Agent Development: Custom ReAct Loop, Tool Registry Design, Tool Calling, Prompt Injection Protection, Prompt Engineering

LLM Integration: Multi-Model Fallback (Groq SDK), LLM Output Validation, tiktoken, Streaming (SSE)

Languages: TypeScript, JavaScript, SQL, Python (learning)

Backend & APIs: Node.js, Express.js, Next.js, REST APIs, SSE, Socket.IO, WebSockets

Database & Cache: PostgreSQL, Prisma, Redis, MongoDB

Validation & Auth: Zod, Clerk, Svix, JWT, bcrypt

Testing: Vitest, Supertest, React Testing Library

Observability: Sentry, OpenTelemetry, Pino

PROJECTS

Gigi AI Ops — Agentic Customer Support Backend

Node.js, Express 5, TypeScript, Groq SDK, Zod, Xenova Transformers, PostgreSQL, Prisma, Clerk, Sentry, Vitest

- Built custom ReAct (Reasoning + Action) loop from scratch — LLM autonomously handles order tracking, cancellations, and refunds across multi-step reasoning cycles without LangChain
- Developed Zod-validated tool registry — AI executes only pre-approved backend functions with schema validation and ownership verification; zero unvalidated tool calls reach the database
- Implemented model fallback routing: Llama 3.3-70B → Llama 3.1-8B on failure — ensures 100% agent availability under Groq API degradation
- Built LLM Ops tracking with tiktoken — logs prompt + completion tokens and calculates real-time session cost in USD/INR per conversation
- Enforced prompt injection protection via strict system prompting and Trusted Backend Context pattern — prevents unauthorized data access and identity spoofing
- Monitored agent performance with Sentry + Pino structured logging — database-level tracking of LLM latency and iteration count per session

github.com/shubhambhattacharya-dev/Agentic_Project | agentic-project-drab.vercel.app

Medo Copilot — Multi-Provider AI Launch Auditor

Next.js, TypeScript, Vercel AI SDK, Playwright, Clerk

- Built multi-provider LLM waterfall fallback — Gemini 2.0 → Groq/Llama 4 Scout → OpenRouter/Claude — automatic failover with zero manual intervention
- Designed hybrid scoring engine: 60% AI-generated insights + 40% deterministic Lighthouse and static analysis metrics — structured JSON output for every audit
- Built static security analyzer scanning for hardcoded secrets and code quality issues — Playwright captures desktop + mobile screenshots for multimodal LLM review

github.com/shubhambhattacharya-dev/Medo_Copilot | medo-copilot.shubhambhattacharya.dev

DocNow — Telemedicine SaaS Platform

Next.js, TypeScript, Clerk, Prisma, PostgreSQL, Vonage

- Built full-stack telemedicine platform with WebRTC video calls, role-based dashboards, and credit-based payment system with full audit trail
- Implemented doctor payout system with platform fee calculation and admin approval workflow

github.com/shubhambhattacharya-dev/health | docnow-zeta.vercel.app

EDUCATION

BCA — Parul University | 2022 — 2025 | CGPA: 8.0/10

Capstone: Gigi AI Ops — production agentic backend with custom ReAct loop and LLM Ops monitoring

Relevant Coursework: Data Structures & Algorithms, DBMS, Computer Networks, Operating Systems, Web Technologies